

Three-dimensional data of wire-cut surface scans under the confocal microscope

Yuhang Lin^{1,2,*}, Heike Hofmann^{2,3}, Curtis Mosher⁴, Eden Amin⁵, Jeff Salyards², Alicia Carriquiry^{1,2}

¹Department of Statistics, Iowa State University, USA

²Center for Statistics and Applications in Forensic Evidence, Iowa State University, USA

³Department of Statistics, University of Nebraska-Lincoln, USA

⁴Roy J. Carver High Resolution Microscopy Facility, Iowa State University, USA

⁵Forensic Science Institute, University of Central Oklahoma, USA

*Corresponding author: Yuhang Lin (yhlin@iastate.edu)

Abstract

Wire-cut striation marks provide evidence for forensic source identification, but publicly available datasets and reproducible quantitative workflows for analyzing these marks are limited. We present a dataset of 120 three-dimensional topographic scans in **x3p** format from aluminum wires cut with five KWS-105 wire cutters. The study contains six scans—two replicates at each of three cutting locations—for each of the 20 tool-edge combinations. We provide additional scan metadata, manually extracted profiles, processed signals, cross-correlation function (CCF) values for pairwise aligned signals, images, and analysis code. The accompanying workflow covers scan inspection, profile and signal extraction, alignment, pairwise similarity visualization, CCF distributions for same-source and different-source comparisons, and variability assessment across replicates, stagings, and acquisitions. The dataset and workflow support the development, evaluation, and reproducible comparison of quantitative methods for wire-cut evidence.

[Agent: Remove the table of contents before preparing the clean Scientific Data submission. The journal’s Data Descriptor outline does not include a table of contents; this one is included only as an internal review aid.]

Contents

Background & Summary	2
Data Acquisition and Processing	3
Scan acquisition	3
Signal extraction	4
Pairwise alignment of signals	5
Data Records	6
Data Overview	9
Technical Validation	10
Usage Notes	15
Data Availability	15
Code Availability	15
References	15

Background & Summary

In forensic analysis, marks left at crime scenes play an important role, and forensic examiners are particularly interested in identifying their origin, a process known as the Source Identification Problem in Forensic Science¹. The forensic science community focuses on two distinct problems: the **specific source problem**, in which examiners aim to determine whether a mark was left by a particular tool, and the **common source problem**, in which examiners aim to determine whether two marks were left by the same tool. Currently, accepted practice for both problems relies on visual inspection of physical items under a comparison microscope. Examiners then report their conclusions according to the Theory of Identification², developed by the Association of Firearm and Toolmark Examiners (AFTE), using one of three categories: *identification*, in which the marks are believed to have been made by the same source; *elimination*, in which the marks are believed to have been made by different tools; and *inconclusive*, in which the similarities or dissimilarities between marks are insufficient to support either identification or elimination. This process has been criticized by the National Research Council (NRC)³ and the President’s Council of Advisors on Science and Technology (PCAST)⁴ for its lack of objectivity and absence of quantifiable measurements, such as error rates.

In discussing error rates, Biedermann⁵ differentiates between internal and external perspectives. The external perspective allows general statements based on black-box studies that relate examiners’ conclusions to ground truth; however, it does not consider the evidence in a particular case. Introducing an internal perspective requires evaluating similarity between pairs of specimens with known ground truth. Algorithms allow these similarities to be quantified objectively and reproducibly.

Advances in microscopy have made it possible to move from optical two-dimensional imaging to digital three-dimensional topographic scanning, allowing algorithmic assessment of the similarity between pairs of forensic pattern evidence⁶. This transition has already supported practical advances in bullet and cartridge comparisons, including visual inspection of bullet line scans from three-dimensional topographic images⁷, the use of features extracted by fast Fourier transform (FFT) as the basis of an artificial intelligence system⁸, an automated pilot study based on consecutively matching striae⁹ (a feature also used by some laboratories in casework), and the incorporation of three-dimensional imaging into routine casework at the Alabama Department of Forensic Sciences (ADFS)¹⁰.

When cut wires are found at a crime scene, understanding the characteristics of marks left by bladed tools becomes essential. When a blade cuts a wire, it leaves striations on the surface, as shown in Figure 1. These striations provide a basis for similarity assessment and source identification. Similar scans have supported the development of quantitative methods for analyzing firearms and other toolmark evidence. For example, studies of screwdriver striations have examined key factors affecting comparisons, including angles of attack¹¹, rotational axes¹², and cutting directions¹³. Foundational research on forensic toolmarks has also emphasized data collection and analysis, including the collection and distribution of datasets for bullet and toolmark analysis^{14,15}, numerical methods for improving accuracy and consistency^{16,17}, and techniques for quantifying similarity^{18,19}.

Data collected at crime scenes lack the known ground truth needed to estimate error rates; therefore, controlled studies are needed. In this study, we provide a publicly available dataset of wire-cut scans together with a systematic quantitative processing and comparison workflow. The shared materials include raw scans, metadata, manually extracted profiles, processed signals, pairwise cross-correlation function (CCF) values, derived images, and analysis code.

The Methods section describes sample collection, scanning, profile extraction, signal processing, and CCF-based alignment. Data Records describes the shared files and repository organization. Data Overview presents descriptive summaries of pairwise CCF values. Technical Validation assesses signal-processing and alignment behavior and variability across replicates, stagings,



Figure 1. Microscopic wire-cut striations. Close-up of striations left by a blade on the cut end of a wire.

and acquisitions. Usage Notes provides practical guidance for working with the shared files, and Code Availability identifies the custom processing and validation code. The dataset and workflow may support future quantitative research on wire-cut evidence.

Data Acquisition and Processing

Five Kaiweets KWS-105 multipurpose tools with 0.5-inch carbon-steel cutting blades were used to cut 16-gauge anodized pure-aluminum wire.

While aluminum is not commonly used for electric wiring because of its poor conductivity, it was selected for its nontoxicity and suitability for receiving cutting marks and retaining them during subsequent handling. Pure aluminum has a Brinell hardness of approximately 15 HB, which places it between lead (approximately 5 HB) and copper (approximately 35 HB).

Scan acquisition

The blades of each of the five wire cutters were labeled A, B, C, and D (see Figure 2 (a)). B opposes A; the reverse sides are labeled D for A and C for B. Three locations on each blade were marked (inner, middle, and outer), as shown in Figure 2 (a). Each four-inch wire segment was cut in half, creating two cut sides, AB and CD. The cut surface of the wire is not flat; under the microscope, it reveals a roof-like structure with a ridge separating the markings made by each blade (see Figure 1). The two roof sides of each cut surface were scanned separately, yielding scans A and B from cut side AB and scans C and D from cut side CD. As an example, the four scans resulting from cut 1 by tool 1 are shown in Figure 2 (c).

The cut surfaces were scanned at 20 $\times$ magnification using a Sensofar confocal microscope, shown in Figure 2 (b). The scans were exported in x3p format (ISO 25178-72:2017/Amd 1:2020) with a lateral resolution of 0.645×0.645 micrometers per pixel and a vertical resolution of 0.1 micrometer.

The complete design therefore consists of 5 tools $\times$ 3 locations $\times$ 2 replicates $\times$ 4 cut surfaces = 120 scans.

The naming of each piece included the specific tool, its edge, location, and replicate according

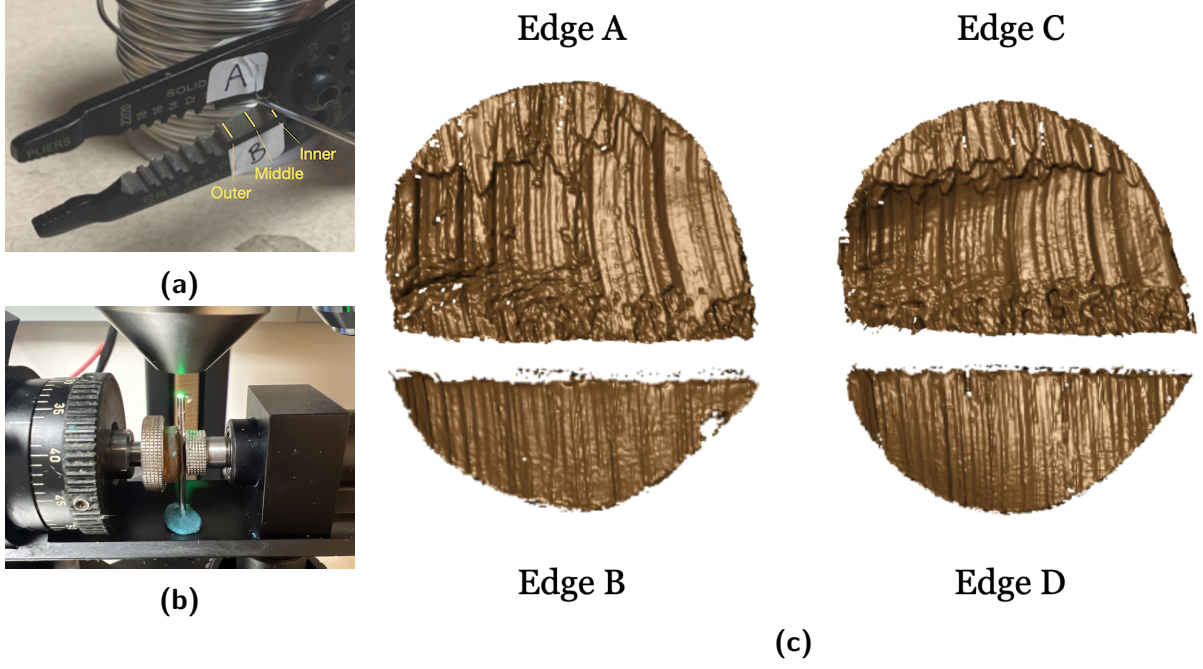


Figure 2. Wire-cut sampling and scanning workflow. (a) A Kaiweets KWS-105 wire cutter with inner, middle, and outer locations marked. (b) A confocal microscope used to scan wire-cut surfaces. (c) Four scans of cutting marks labeled according to the corresponding cutting edges A, B, C, and D.

to the following convention:

```

T__W-L_-R_
||  |  |
||  |  +-- Replicate: 1 or 2
||  +----- Location: I (inner), M (middle), or O (outer)
|+----- Edge: A, B, C, or D
+----- Tool: 1, 2, 3, 4, or 5

```

For example, T1AW-LI-R1 denotes the first replicate of a piece cut with edge A of tool 1 at the inner location.

To minimize errors during data collection, two team members cut and labeled the wire pieces together. One team member scanned the wire cuts and assigned names to the scans as described.

Signal extraction

Signals were extracted from each surface scan using a two-step procedure. First, a representative profile was identified along a line across the surface perpendicular to the dominant striation direction. This line was chosen manually to avoid scanning artifacts (e.g., holes) and cutting artifacts (e.g., rips caused by rolling the wire). Along this line, surface values were assigned at equally spaced positions using the observed surface value at the nearest position on the surface grid.

In a second step, we extract the scan’s signal by removing trends and outliers from the surface values of the profile with the help of two LOWESS filters with bandwidths of $400\mu m$ and $40\mu m$, respectively²⁰. All smoothers are susceptible to boundary issues, particularly when outliers are present. Steep changes at the end of a signal have a high potential to attract another signal during alignment, causing misalignments. To avoid potential artificial “hooks” at either end of a signal, two-sample t -tests were applied to the surface values in the outermost 2.5% intervals along the horizontal axis. Specifically, the summary statistics (sample size, mean, and

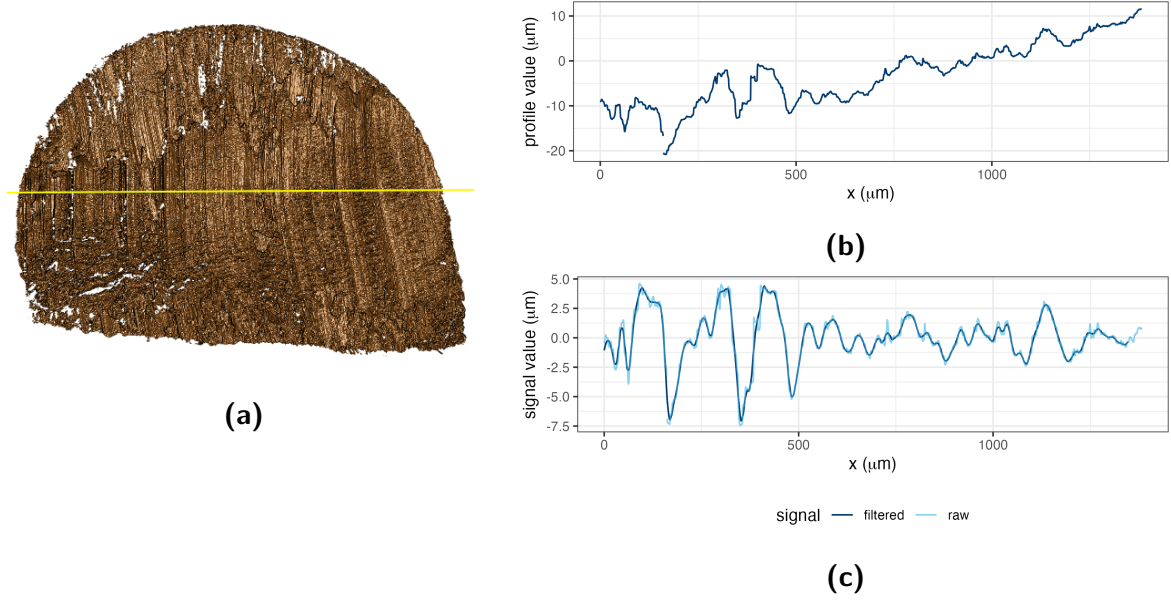


Figure 3. Profile and signal extraction. (a) A dark-blue profile line drawn across the scan striations. (b) The profile function extracted along the line in (a). (c) Raw and filtered signals obtained from the profile function in (b).

standard deviation) of the signal values within the intervals $[0, 2.5]\%$, $[2.5, 5]\%$, $(95, 97.5]\%$, and $(97.5, 100]\%$ were calculated. At each end, the test compared the outermost 2.5% interval with the adjacent interval. The signal values in the outermost interval were retained only when the corresponding p -value was above 0.01, indicating no significant shift.

The widths of these intervals correspond to the choice of bandwidths and the length of the signal. The interval widths were determined using cross-validation.

The resulting signals are used for subsequent comparisons with other signals.

Figure 3 shows a scan on the left (a) with the location of the manually chosen profile line in yellow. The corresponding profile and signal are plotted in (b) and (c).

One profile for each scan is included in folder `profiles/` as a csv file in the repository using the same naming convention as the scans, with the precursor of `profiles-`. More details on the exact format of these files are provided below.

The code to create signals from profiles is available in `code/4-create_signals_from_profiles.R` and a single file containing all 120 signals is available as `data-derived/wire-signals.csv.zip`.

Pairwise alignment of signals

Before signals from different scans can be compared, they need to be aligned. For each pair of signals, the value of the cross-correlation function (CCF) is calculated across relative shifts. The shift corresponding to the maximum CCF is retained as the alignment position, while the maximized cross-correlation function, CCF_{max} , gives a measure of the similarity of the signals, with a value of 1 indicating complete similarity (apart from linear scaling) and a value of 0 indicating (statistical) independence.

Figure 4 shows all signals corresponding to cuts 1 and 2 made with tool 1 at the location closest to the pivot area (inner location). Each of the four edges corresponds to one row in the figure. The lines in the first two columns correspond to the extracted signals for cuts 1 and 2, respectively. The third column shows the two signals aligned at the location of the maximal CCF. The cross-correlation value for each pair is included as part of the title of the chart.

The code to align signals pairwise is available in `code/6-align-pairwise.R`, and `data-derived/wire_pairwise_ccf.csv` contains the CCF similarity scores of all pairwise aligned

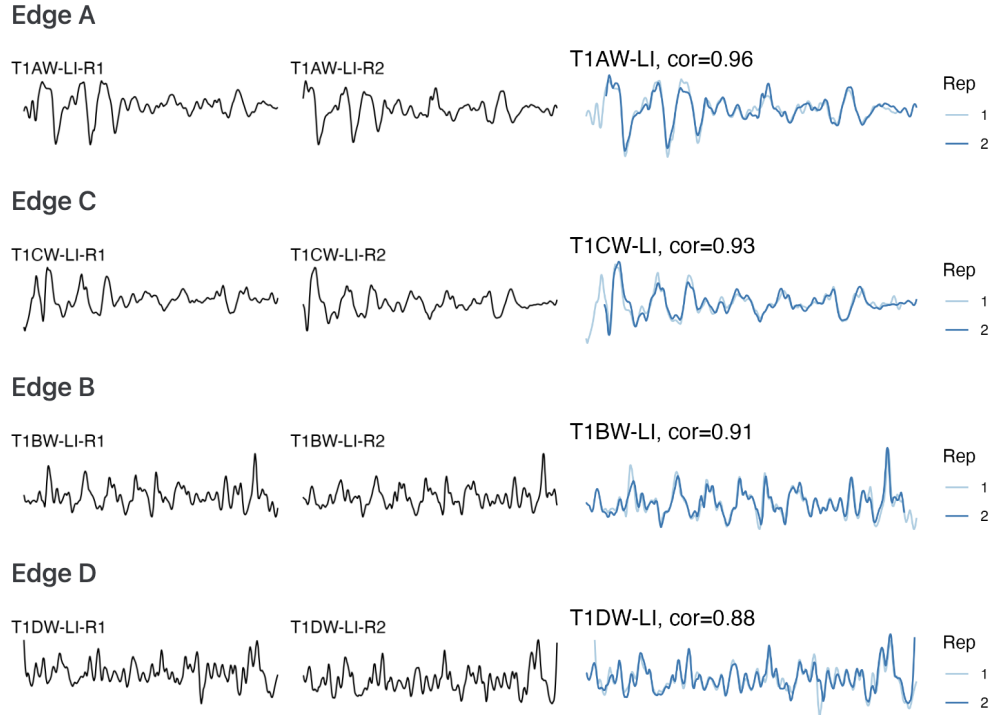


Figure 4. Same-source signal alignment. Rows correspond to cut surfaces A, C, B, and D. The first two columns show signals extracted from scans corresponding to the replicate cuts of tool 1 at location I; the third column overlays the aligned signals and reports their CCF values.

signals.

Data Records

The repository record on Iowa State University DataShare [DATASET CITATION] at [https://doi.org/\[DOI\]](https://doi.org/[DOI]) contains the x3p scans, metadata, extracted profiles, processed signals, pairwise CCF values, and derived images. [Agent: Replace the dataset citation and DOI placeholders. For first-round review, the destination must permit anonymous download; from the second round onward, it must be a public formal repository record. Add the full dataset reference to References.]

The repository contents, formats, and folder organization are summarized in Table 1.

A data dictionary defining non-obvious columns in `meta.csv`, the profile and signal files, and the pairwise-CCF file is provided in [repository documentation path].

The dataset is available under the [license] license. [Agent: Verify Table 1 against the deposited record and replace the documentation-path and data-license placeholders. For this non-sensitive original dataset, use CC0 or CC BY rather than a license containing NC or SA restrictions.]

[Heike: We need to include a descriptor table for each of the csv files; I started with `meta.csv` - the formatting is still horrific]

The draft variable descriptions for `meta.csv` are listed in Table 2.

Table 2. Structure of `meta.csv`.

Variable	Type	Description
x3p_path	text	We need to change those paths to scans/T...x3p
tool	integer	Tool used, one of 1 through 5

Variable	Type	Description
edge	character	Cutting edge, one of A, B, C, or D
cut	delete	HH: cut number - we don't need that column anymore
scan_no	delete	HH: scan number - remove this column
location	text	Cutting location, one of Inner, Middle, or Outer
replicate	integer	Replicate, one of 1 or 2 for the general study, higher values are used for validation
ID	text	identifier for each of the cuts as specified in ...
height	integer	height of the scan in pixels
width	integer	width of the scan in pixels

187 The draft variable descriptions for the profile CSV files are listed in Table 3.

Table 3. Structure of any of the `profile-....csv` files.

Variable	Type	Description
x	numeric	horizontal coordinate in micrometers, multiple of 0.645
value	numeric	observed surface height
y	numeric	y coordinate, constant, set to zero
src	text	source cut

188 The draft variable descriptions for `wire-signals.csv` are listed in Table 4.

Table 4. Structure of the `wire-signals.csv` file.

Variable	Type	Description
extrct_profiles	delete	NA
id	text	identifier
tooledge	text	combination of tool and edge
location	text	cut location, one of Inner, Middle, or Outer
rep	integer	Replicate, either 1 or 2 for the regular study, higher values are used for variance assessment
tool	integer	Cutting tool, between 1 and 5
edge	character	Cutting edge, one of A, B, C, or D
x	numeric	Horizontal location in micrometers
value	numeric	Observed surface location in micrometers
y	numeric	y location; constant
n	delete	NA
detrend	numeric	Detrended surface location
raw_sig	numeric	HH: hook processed?
sig	numeric	Processed signal value

189 The draft descriptor for `wire_pairwise_ccf.csv` is reproduced in Table 5.

Table 5. Structure of the `wire_pairwise_ccf.csv` file.

Variable	Description
x	horizontal coordinate in micrometers, multiple of 0.645
value	observed surface height

Variable	Description
y	y coordinate, constant, set to zero
src	source cut

190 [Agent: The recovered draft table for `wire_pairwise_ccf.csv` repeats the profile
191 fields rather than describing pair identifiers and CCF scores. Confirm the columns
192 and their definitions for the pairwise-CCF file before replacing these draft rows.]

Table 1. Structure of available data and files.

Data type	Description	Section
Raw data		
scans/	folder containing 120 topographic 3D scans corresponding to 30 aluminum wire cuts (x3p format)	Scan acquisition
meta.csv	metadata for each cut, including tool, blade, and location information (CSV format)	Scan acquisition
Manual derivatives		
profiles/	folder containing manually extracted profiles (CSV format)	Signal Extraction
Computational derivatives in the folder 'data-derived/'		
wire-signals	signals processed from the corresponding profiles (zipped CSV format)	Signal Extraction
wire_pairwise_ccf	CCF values for all pairwise aligned signals (zipped CSV format)	Signal Alignment
Image files		
pngs/	folder containing images of 3D wire-cut scans (PNG format)	Scan acquisition
profile-images/	folder containing images of profiles extracted from wire cuts (PNG format)	Signal Extraction
Visual inventory in the folder 'assessment/analysis-manual/'		
processing-wires	display of pairwise aligned signals from the same sources (HTML format)	Signal Alignment
Technical validation		
variability-assessment/	folder containing technical-validation documents, with an internal structure that mirrors the main folder	Technical Validation

Data Overview

As descriptive summaries of comparison behavior across the dataset, Figure 5 displays pairwise CCF values organized by tool, cut surface, blade location, and replicate. The main diagonal contains self-comparisons, whereas the off-diagonal cells compare distinct signals. Figure 6 displays the CCF distributions for same-source and different-source comparisons.

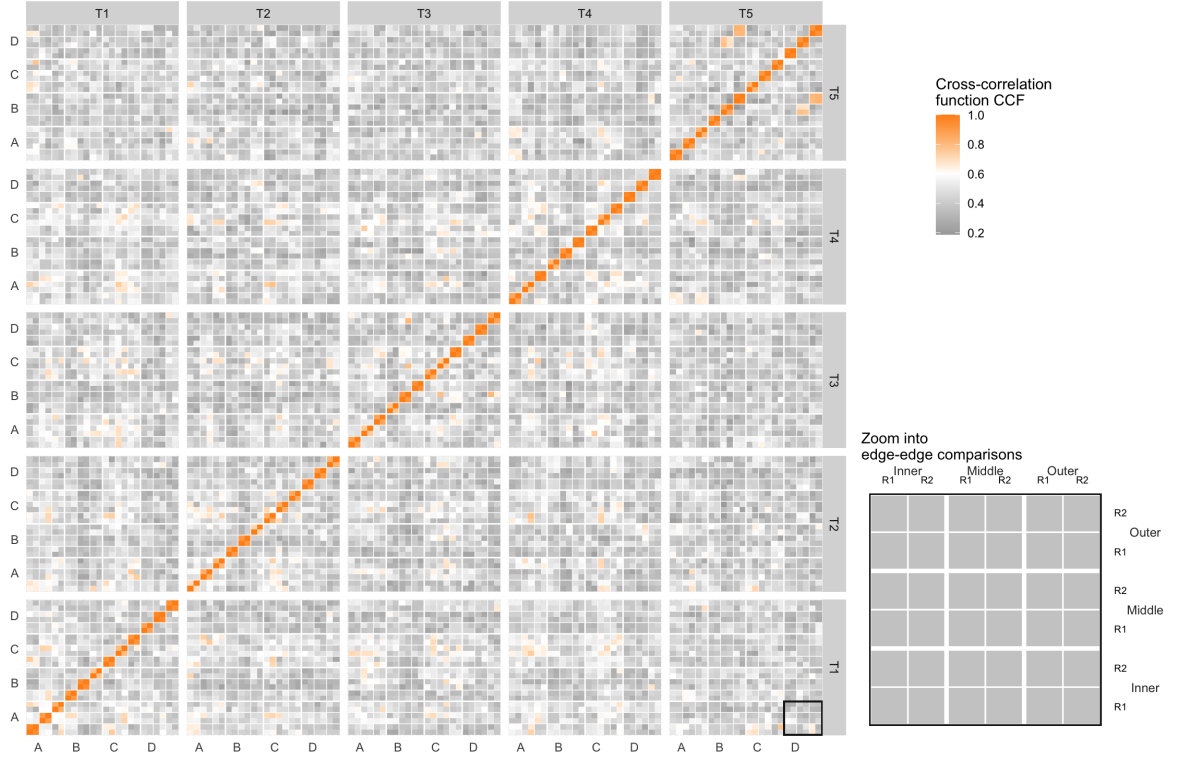


Figure 5. Pairwise CCF tile map. Rows and columns group scans by tool, cut surface, blade location, and replicate. Orange cells indicate larger CCF values and gray cells indicate smaller values; the inset enlarges the boxed edge-to-edge comparison block.

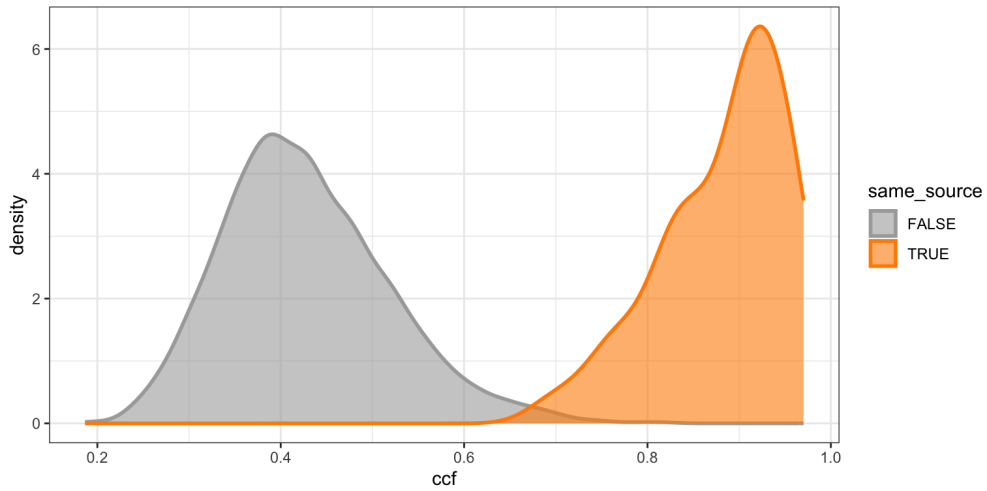


Figure 6. CCF density distributions. Kernel density estimates are shown for different-source comparisons in gray and same-source comparisons in orange.

Technical Validation

Technical validation was conducted in two areas: (1) signal processing and alignment and (2) variability assessment.

Scan processing and signal alignment were validated as follows. High CCF values were expected between signals from the same source and low values between signals from different sources. Figure 4, shown earlier in Align signals, meets the expected pattern for same-source signals.

204 As an example of a different-source comparison, T1AW-LI-R1 and T2AW-LI-R1 were selected.
 205 Figure 7 (a) and (b) show the selected scans. Their signal alignment, shown in Figure 7 (c),
 206 has a CCF value of 0.2, which is as low as expected. For the following summaries, edges A
 207 and C were categorized as long, edges B and D as short, and comparisons pairing one long
 208 edge with one short edge as mixed. The resulting CCFs for all pairwise comparisons were
 209 also summarized using boxplots and receiver operating characteristic (ROC) curves. Figure 8
 210 and Figure 9 show these summaries. The ROC curves plot sensitivity, or the true positive rate,
 211 against the false positive rate (FPR; 1 - specificity). The curves are close to the upper-left corner,
 212 indicating strong classification performance. For the combined comparison set, a threshold of
 213 0.589 controls the FPR below 0.05 with a false negative rate (FNR) of 0, whereas a threshold
 214 of 0.658 controls the FPR below 0.01 with an FNR of 0.02. These results are consistent with
 215 the expected pattern and validate the pipeline for processing and aligning signals.

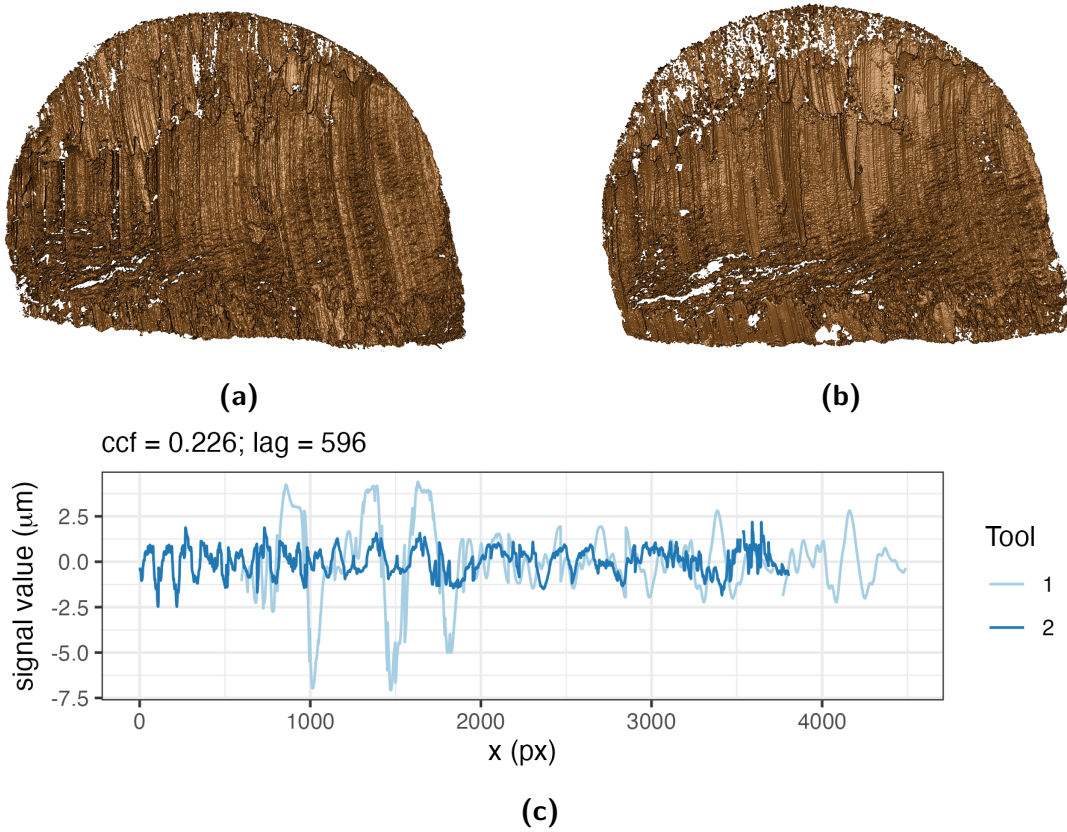


Figure 7. Different-source scan and signal comparison. (a) Scan T1AW-LI-R1 cut by tool 1. (b) Scan T2AW-LI-R1 cut by tool 2. (c) Alignment of signals from T1AW-LI-R1 and T2AW-LI-R1.

216 The second validation assessed variability in the pipeline across replicates, stagings, and acqui-
 217 sitions. Scans from tool 2 at the middle location were selected for this assessment. Replicate
 218 variability was assessed using two different cuts at the same edge and position. The comparison
 219 in Figure 4 in Align signals illustrates this assessment. To assess staging variability, replicate 2
 220 was rescanned under the confocal microscope, and the staging label S was introduced into the
 221 naming protocol. The S1 designation is omitted from labels, so R2 is equivalent to R2-S1. For
 222 all four edges of tool 2 at the middle location, three stagings (R2, R2-S2, and R2-S3) were cre-
 223 ated, pairwise CCFs between stagings were computed, and their mean and standard deviation
 224 (SD) were calculated.
 225 Within each staging, the effect of different acquisitions was further examined by keeping the
 226 samples on the confocal microscope while varying the lighting conditions. A was introduced
 227 as the acquisition label. For edge A at staging 3, four acquisitions were performed, and the

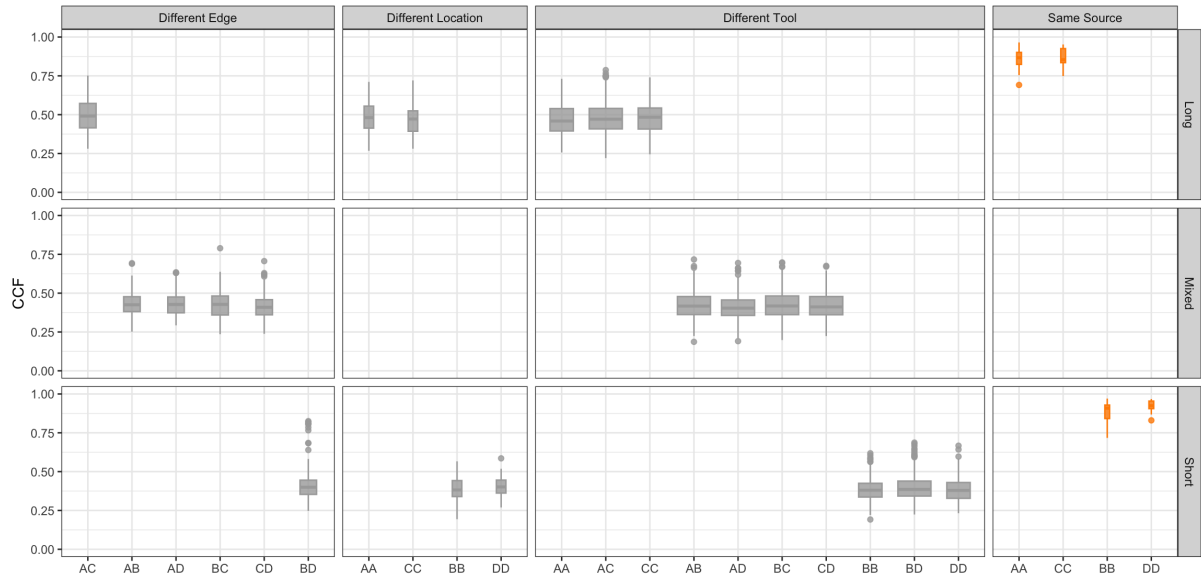


Figure 8. CCF distributions across comparison classes. Gray boxplots show different-source comparisons grouped by different edge, location, or tool; orange boxplots show same-source comparisons. Rows distinguish long, mixed, and short edge comparisons.

A1 designation was omitted from the first acquisition label. Therefore, R2-S3 is equivalent to R2-S3-A1. For the other three edges, two acquisitions were conducted at staging 3. These acquisitions are denoted by R2-S3 and R2-S3-A2. Within staging 3, pairwise CCFs between acquisitions were computed, and their mean and standard deviation were calculated. Variability between distinct stagings was assessed separately through the staging comparisons described above.

With this setup, CCF values were expected to become more consistent from replicate to staging to acquisition as the scanning conditions became progressively more stable. Consequently, variability was expected to decrease along this sequence. Replicates, stagings, and acquisitions were also compared, with smaller differences expected between acquisitions and stagings than between stagings and replicates. The results were visualized as scatter plots. Figure 10 shows the scatter plots and intervals extending two standard deviations above and below the mean. Higher orange mean lines indicate an increase in the mean CCF, and narrower shaded intervals indicate a decrease in the standard deviation. The detailed numerical results, including means, standard deviations, and differences between means, are shown in Table 6 and confirm the expected pattern.

[Agent: The manuscript contains 11 figures. Scientific Data recommends no more than eight but does not mandate this limit. Confirm with the coauthors or professor whether any logically connected pairs should be combined for revised submission; do not remove a figure or its discussion without an author decision.]

The variability assessment does not include variation in CCF values caused by differences in the manual extraction of profiles from scans. To support reproduction of the results reported here, the extraction locations and values for each scan have been included.

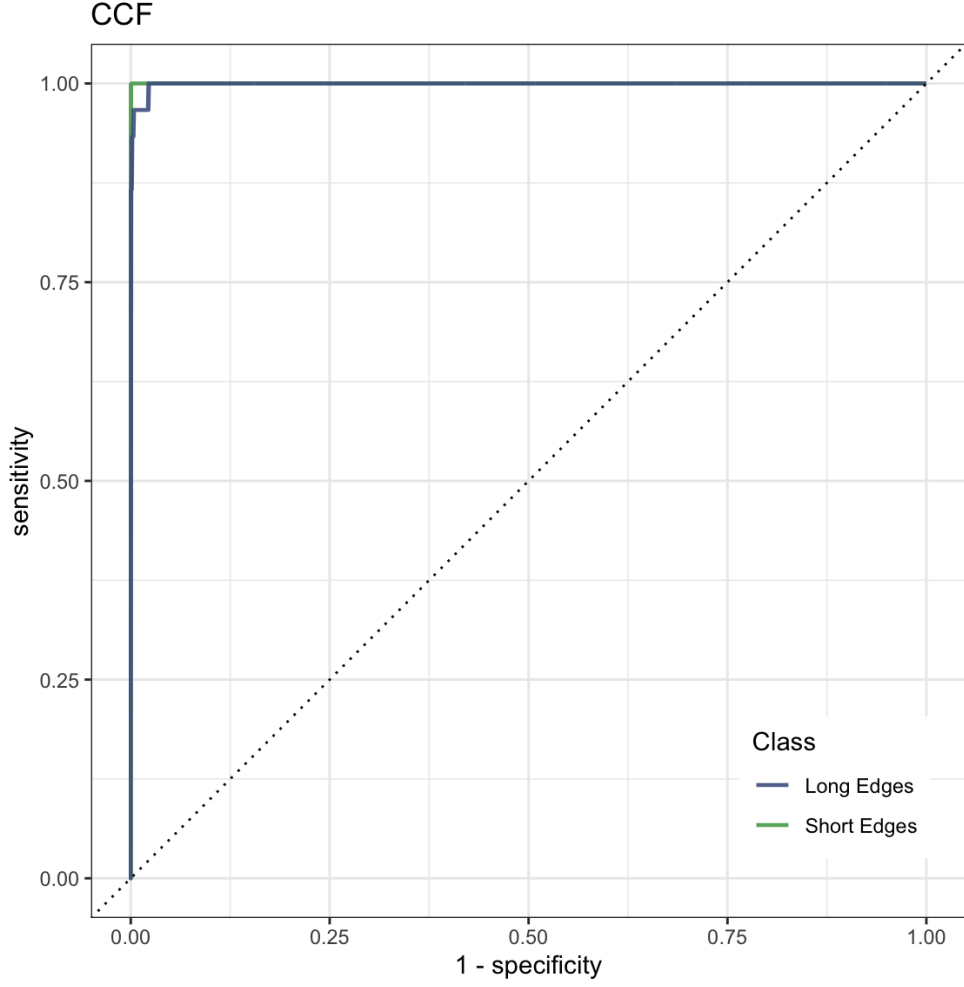


Figure 9. ROC curves for CCF-based classification. Curves are shown separately for long and short edge classes; the dotted diagonal indicates chance-level classification.

Table 6. Comparison of CCF means and standard deviations across different replicates, stagings, and acquisition settings. Dashes indicate that the standard deviation is undefined because only one CCF value is available.

Edge	Replicate (R)		Staging (S)			Acquisition (A)		
	Avg_R	SD_R	Avg_S	SD_S	$\text{Avg}_S - \text{Avg}_R$	Avg_A	SD_A	$\text{Avg}_A - \text{Avg}_S$
A	0.818	0.015	0.961	0.012	0.143	0.992	0.003	0.031
B	0.879	0.031	0.922	0.042	0.043	0.939	—	0.017
C	0.841	0.022	0.952	0.021	0.111	0.959	—	0.007
D	0.797	0.011	0.966	0.017	0.169	0.986	—	0.020

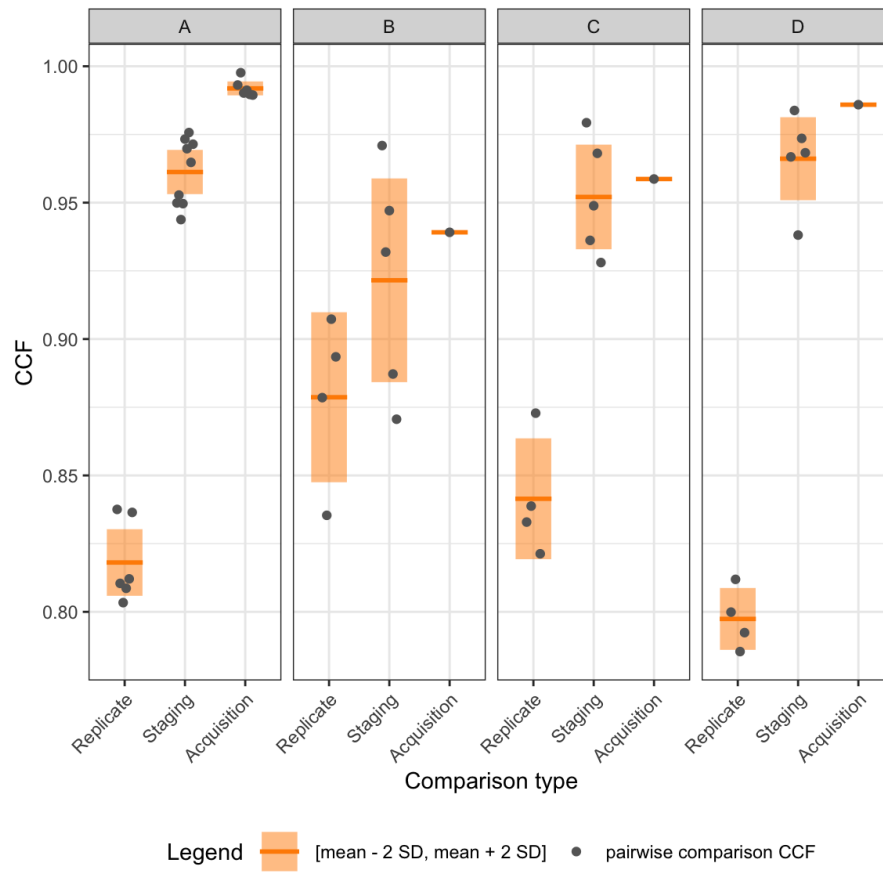


Figure 10. Variability across scanning conditions. Panels A, B, C, and D correspond to the four cut surfaces. Gray points show pairwise CCF values, orange horizontal lines show means, and shaded intervals extend two standard deviations above and below each mean.

Usage Notes

The raw surface scans are provided in **x3p** format. To inspect or process these files in R, use the **x3ptools** package and its reference manual, available from CRAN²¹. The **meta.csv** file links each scan to its tool, edge, blade location, and replicate identifiers. The supplied profiles and processed signals provide a starting point for users who do not wish to repeat manual profile extraction. The supplied extraction coordinates allow these derived files to be related back to the corresponding scans. Because each supplied signal is based on a manually selected profile, users assessing sensitivity to profile placement should re-extract profiles from the raw **x3p** scans using alternative documented locations.

Data Availability

The data described in this Data Descriptor are available from Iowa State University DataShare [DATASET CITATION] at [https://doi.org/\[DOI\]](https://doi.org/[DOI]). [Agent: Replace the dataset citation and DOI placeholders with the same record used in Data Records.]

Code Availability

Custom R code was used to inspect raw scans, extract profiles, derive signals, align signals, and generate the validation outputs. The code is available in the **assessment/code** folder at <https://github.com/heike/wirecuts-data>. [Agent: For permanent access, archive the exact code release in a DOI-providing repository and cite it if feasible; Scientific Data recommends combining GitHub with a DOI archive.] The files and their roles are described in Table 7.

The provided code reproduces the processing, comparison, validation, and visualization outputs described in Data Acquisition and Processing and Technical Validation. The workflow was run using R [version] and **x3ptools** [version]. Versions of additional R packages are recorded in [environment file or repository documentation path]. [Agent: Replace the R, **x3ptools**, and environment-record placeholders after checking the analysis environment.]

Table 7. Overview of available code.

Code	Description	Section
Inspect raw scans		
<code>1-create_pngs_from_x3p.R</code>	obtain images of <code>x3p</code> files in <code>scans/</code>	Cutting wires
Extract profiles		
<code>2-create_profiles_from_x3p.R</code>	manually extract profiles from each scan	Extract profiles
<code>3-create-single-profile-file.R</code>	create meta profile information	Extract profiles
Derive signals		
<code>4-create_signals_from_profiles.R</code>	derive signals from each profile	Filtered signals
Align signals		
<code>5-create-images.R</code>	create images for pairwise alignment	Align signals
<code>6-align-pairwise.R</code>	compute pairwise alignment CCF values	Align signals
<code>7-all-comparison-results.R</code>	visualize comparison results	Align signals

References

- [1] Lindley, D. V. A Problem in Forensic Science. *Biometrika* **64**, 207–213, 10.2307/2335686 (1977).
- [2] AFTE. The association of firearm and tool mark examiners: Theory of identification as it relates to toolmarks. *AFTE Journal* **30**, 86–88 (1998).
- [3] National Research Council (U.S.). *Strengthening Forensic Science in the United States: A Path Forward* (National Academies Press, 2009).
- [4] President’s Council of Advisors on Science and Technology. *Forensic Science in Criminal Courts: Ensuring Scientific Validity of Feature-Comparison Methods* (Executive Office of the President of the United States, President’s Council, Washington, D.C., 2016).
- [5] Biedermann, A. The strange persistence of (source) “identification” claims in forensic literature through descriptivism, diagnosticism and machinism. *Forensic Science International: Synergy* **4**, 15 pages, 10.1016/j.fsisy.2022.100222 (2022).
- [6] Bonfanti, M. S. & Ghauharali, R. I. Visualisation by confocal microscopy of traces on bullets and cartridge cases. *Science & Justice* **40**, 241–256, 10.1016/S1355-0306(00)71994-0 (2000).
- [7] Smith, C. L. Linescan imaging of ballistics projectile markings for identification. In *Proceedings. 36th Annual 2002 International Carnahan Conference on Security Technology*, 216–222, 10.1109/CCST.2002.1049254 (2002).
- [8] Li, D. Ballistics Projectile Image Analysis for Firearm Identification. *IEEE Transactions on Image Processing* **15**, 2857–2865, 10.1109/TIP.2006.877530 (2006).

- [9] Chu, W. *et al.* Pilot Study of Automated Bullet Signature Identification Based on Topography Measurements and Correlations*†. *Journal of Forensic Sciences* **55**, 341–347, 10.1111/j.1556-4029.2009.01276.x (2010).
- [10] McClarin, D. Adding an objective component to routine casework: Use of confocal microscopy for the analysis of 9mm caliber bullets. *AFTE Journal* **47**, 161–170 (2015).
- [11] Baiker, M., Pieterman, R. & Zoon, P. Toolmark variability and quality depending on the fundamental parameters: Angle of attack, toolmark depth and substrate material. *Forensic Science International* **251**, 40–49, 10.1016/j.forsciint.2015.03.003 (2015).
- [12] Garcia, D. L., Pieterman, R. & Baiker, M. Influence of the axial rotation angle on tool mark striations. *Forensic Science International* **279**, 203–218, 10.1016/j.forsciint.2017.08.021 (2017).
- [13] Cuellar, M., Gao, S. & Hofmann, H. An algorithm for forensic toolmark comparisons. *Forensic Science International: Synergy* **9**, 100543, 10.1016/j.fsisyn.2024.100543 (2024).
- [14] Ma, L. *et al.* NIST Bullet Signature Measurement System for RM (Reference Material) 8240 Standard Bullets. *Journal of Forensic Sciences* **49**, 1–11, 10.1520/JFS2003384 (2004).
- [15] Zheng, X. A., Soons, J. A. & Thompson, R. M. NIST Ballistics Toolmark Research Database | NIST (2016).
- [16] Chu, W., Thompson, R. M., Song, J. & Vorburger, T. V. Automatic identification of bullet signatures based on consecutive matching striae (CMS) criteria. *Forensic Science International* **231**, 137–141, 10.1016/j.forsciint.2013.04.025 (2013).
- [17] Vorburger, T. *et al.* Applications of cross-correlation functions. *Wear* **271**, 529–533, 10.1016/j.wear.2010.03.030 (2011).
- [18] Hare, E., Hofmann, H., Carriquiry, A. *et al.* Automatic matching of bullet land impressions. *The Annals of Applied Statistics* **11**, 2332–2356, 10.1214/17-AOAS1080 (2017).
- [19] Ju, W. & Hofmann, H. An Open-Source Implementation of the CMPS Algorithm for Assessing Similarity of Bullets. *The R Journal* **14**, 267–285, 10.32614/RJ-2022-035 (2022).
- [20] Cleveland, W. S., Grosse, E. & Shyu, W. M. Local Regression Models. In *Statistical Models in S* (Routledge, 1992).
- [21] Hofmann, H., Vanderplas, S., Krishnan, G. & Hare, E. "x3ptools: Tools for Working with 3D Surface Measurements", 10.32614/CRAN.package.x3ptools (2018).

Author Contributions

[Agent: Confirm whether all authors approve conversion of this statement to the CRediT taxonomy before revising the contributor roles.]

Y.L.: Methodology, Software, Validation, Data Curation, Writing: Draft; H.H.: Conceptualization, Methodology, Validation, Writing: Review and Editing; C.M.: Laboratory Supervision; E.A.: Physical Specimen, Scanning; J.S.: Forensic Advice; A.C.: Funding Acquisition.

All authors reviewed the manuscript.

Competing Interests

H.H. is a technical advisor to the Association of Firearm and Toolmark Examiners (AFTE), a fellow of the American Statistical Association (ASA), and a member of the ASA Forensic Science Committee. H.H. has testified as a court witness on behalf of Judge April Neubauer in the New York State Supreme Court, Criminal Term, in New York City. [Agent: Confirm that the declaration covers every author and replace Alicia's declaration, if any.] The authors declare the following competing interests:

Funding

This work was partially funded by the Center for Statistics and Applications in Forensic Evidence (CSAFE) through Cooperative Agreement 70NANB20H019 between NIST and Iowa State University. The agreement includes activities carried out at Carnegie Mellon University, Duke University, the University of California, Irvine, the University of Virginia, West Virginia University, the University of Pennsylvania, Swarthmore College, and the University of Nebraska-Lincoln.